

Exercises

Task 1:

Answer the following questions. Additionally, think of how you could explain each topic to someone outside of the course who doesn't study data science.

1. What is the difference between AI, ML, and deep learning?
2. What is machine learning?
3. What is the primary difference between unsupervised and supervised machine learning?
4. What is reinforcement learning?
5. What types of problems can machine learning solve well when compared to traditional programming?
6. What types of problems can you solve with unsupervised machine learning?
7. What types of problems can you solve with supervised machine learning?
8. Explain the difference between a ML algorithm and a ML model?
9. List some of the challenges of machine learning and provide concrete examples?

Task 2:

Please set up a Python environment and all required and essential Python libraries for machine learning on your personal computer.

Task 3:

PCA Iris dataset:

1. Load (all features, using Pandas), standardize this d -dimension dataset (d is number of features) and Split the Iris dataset to training and test sets with ratio 70% and 30%, respectively.
2. Write your own function to calculate the covariance matrix. Then compute the eigenvalues and eigenvectors of this matrix.

3. Plot the cumulative variance ratio (c.f. block [11] of the nbviewer of Chapter 5).
4. Choose the $k = 3$ eigenvectors that correspond to the k largest eigenvalues to construct a $d \times k$ -dimensional transformation matrix $\mathbf{W}$; the eigenvectors are the columns of this matrix.
5. Project the samples onto the new feature subspace, and plot the projected data using the transformation matrix $\mathbf{W}$ (c.f. block [14] of the nbviewer of Chapter 5)

Task 4:

LDA for Iris:

Repeat the steps above PCA but using the matrix $\mathbf{S}_W^{-1}\mathbf{S}_B$ instead of covariance matrix.